

Probabilistic Machine Learning

Lecture 04: Naive Bayes and the Generative Route

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Where We Left Off

Lecture 03 took two shortcuts and named what each drops:

Shortcut	Instead of the ideal	What is dropped
Keep the peak	the most probable parameter (MLE, MAP)	the spread: how sure we are
Only what you need	model $p(y \mid x)$ directly — logistic regression	$p(x)$: every question about the inputs

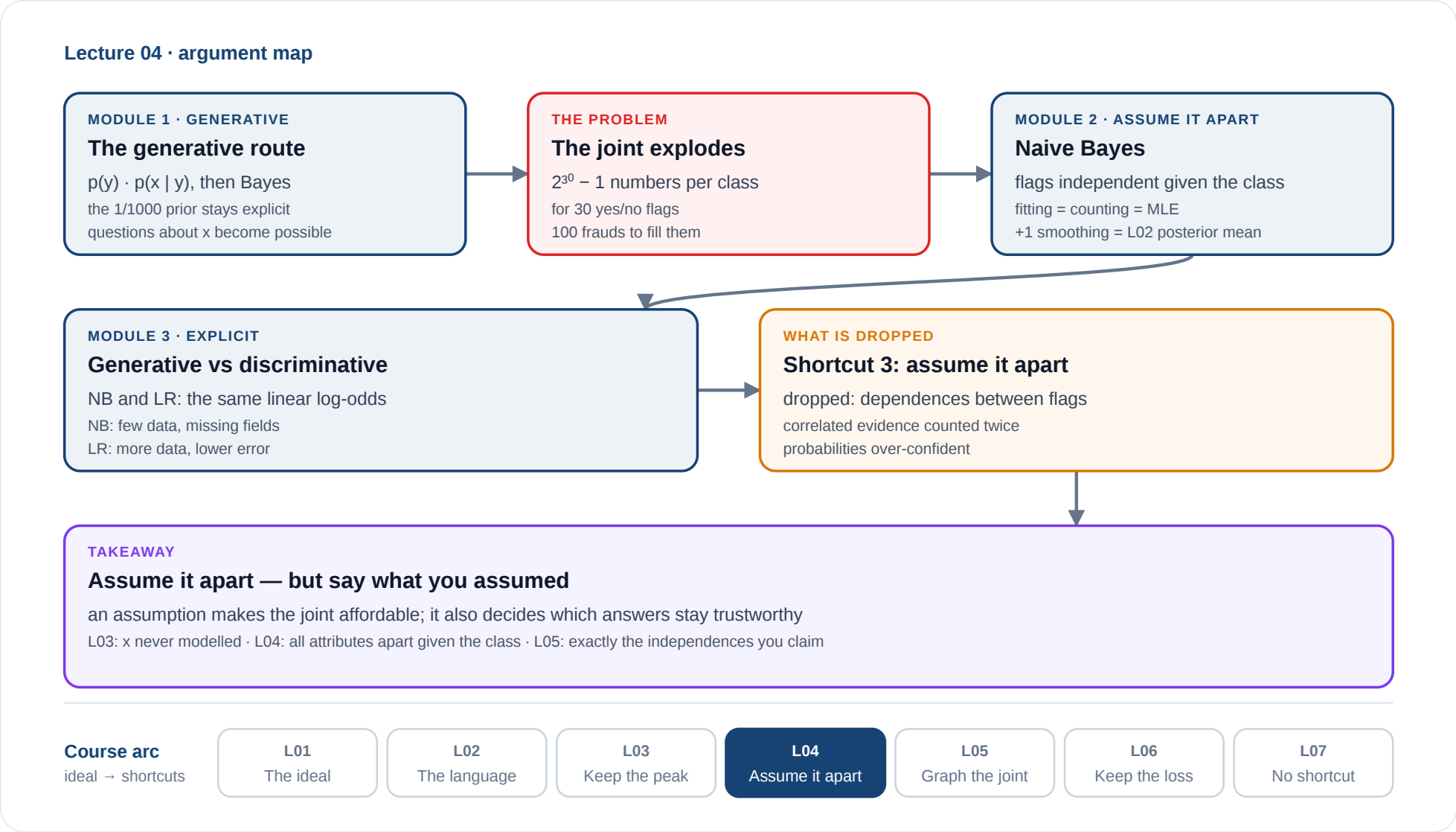
Today the inputs come back. The **generative route** models how the data arises, class by class, and converts with Bayes:

$$p(x, y) = p(y) \cdot p(x \mid y)$$

THE CATCH

$p(x \mid y)$ is a joint over every attribute — and it explodes. Paying for it is **shortcut 3: assume it apart.**

Argument Map



The Generative Route

Guiding Question: Logistic regression never asked what a fraudulent transaction *looks like*. What would it take to model that — and why does nobody do it in full?

Two Routes to the Same Posterior

A labelled table of transactions supports **both** routes to $P(\text{fraud} \mid x)$, where x is the vector of red flags:

DISCRIMINATIVE — LECTURE 03

Model $p(y \mid x)$ directly

Logistic regression: a score for each x .
Nothing is said about how x arises.

GENERATIVE — TODAY

Model $p(y)$ and $p(x \mid y)$

How does a *fraudulent* transaction look, and how an honest one? Then convert with Bayes.

$$P(y \mid x) = \frac{p(x \mid y) \cdot P(y)}{\sum_{y'} p(x \mid y') P(y')}$$

Why Model How Fraud Looks?

Logistic regression already gives $P(\text{fraud} \mid x)$. What can a model of $p(x \mid y)$ do that it cannot?

Why Model How Fraud Looks?

Logistic regression already gives $P(\text{fraud} \mid x)$. What can a model of $p(x \mid y)$ do that it cannot?

- ✓ **Keep the prior explicit.** The 1 in 1,000 of Lecture 02 is a separate factor: a new market with 1 in 200 changes $P(y)$, not the whole model.
- ✓ **Handle a missing field.** An attribute nobody observed can be summed out of $p(x \mid y)$.
- ✓ **Notice the strange.** A transaction with tiny $p(x)$ under *both* classes is unlike anything seen — a score alone cannot say so.

The price: $p(x \mid y)$ is a **joint distribution over all attributes**, one per class.

The Joint Explodes

30

yes/no red flags per transaction



$2^{30} - 1$

≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone



100

frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

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15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

The generative route models **how the data arises** — $p(y)$ and $p(x | y)$ — and converts with Bayes. It keeps the prior explicit and can answer questions about x itself, but it must pay for a **joint over all attributes**, which explodes.

Naive Bayes: Assume It Apart

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

Definition. A and B are **conditionally independent given C** , written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \iff P(A \mid B \cap C) = P(A \mid C)$$

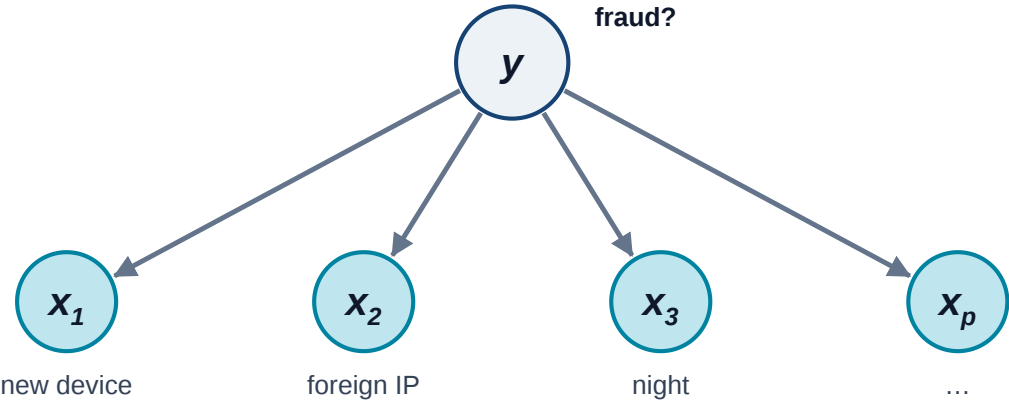
Reading: **once C is known, B tells us nothing more about A .**

Among frauds only: does a new device make a foreign IP more likely?

The Naive Assumption

Assume the attributes are **independent given the class**:

$$p(x_1, \dots, x_p \mid y) = \prod_{j=1}^p p(x_j \mid y)$$



$$p(y) \cdot p(x_1 \mid y) \cdot p(x_2 \mid y) \cdot \dots \cdot p(x_p \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

Shortcut 3 — assume it apart. Dropped: every dependence *within* a class.

Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \quad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v, y)}{n(y)}$$

Lecture 02

Each attribute, within one class, is a **categorical** variable — for a yes/no flag, a Bernoulli.

Lecture 03

Its maximum likelihood estimate is the **relative frequency** — the same derivation as h/n for the coin.

Naive Bayes is fitted in one pass over the table, with no optimiser — and every number it uses is a **peak**. Shortcut 1 is taken as well.

The Classifier

$$P(y \mid x_1, \dots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^p \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio LR}_j}$$

Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds 1 : 999.

flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

Three Red Flags

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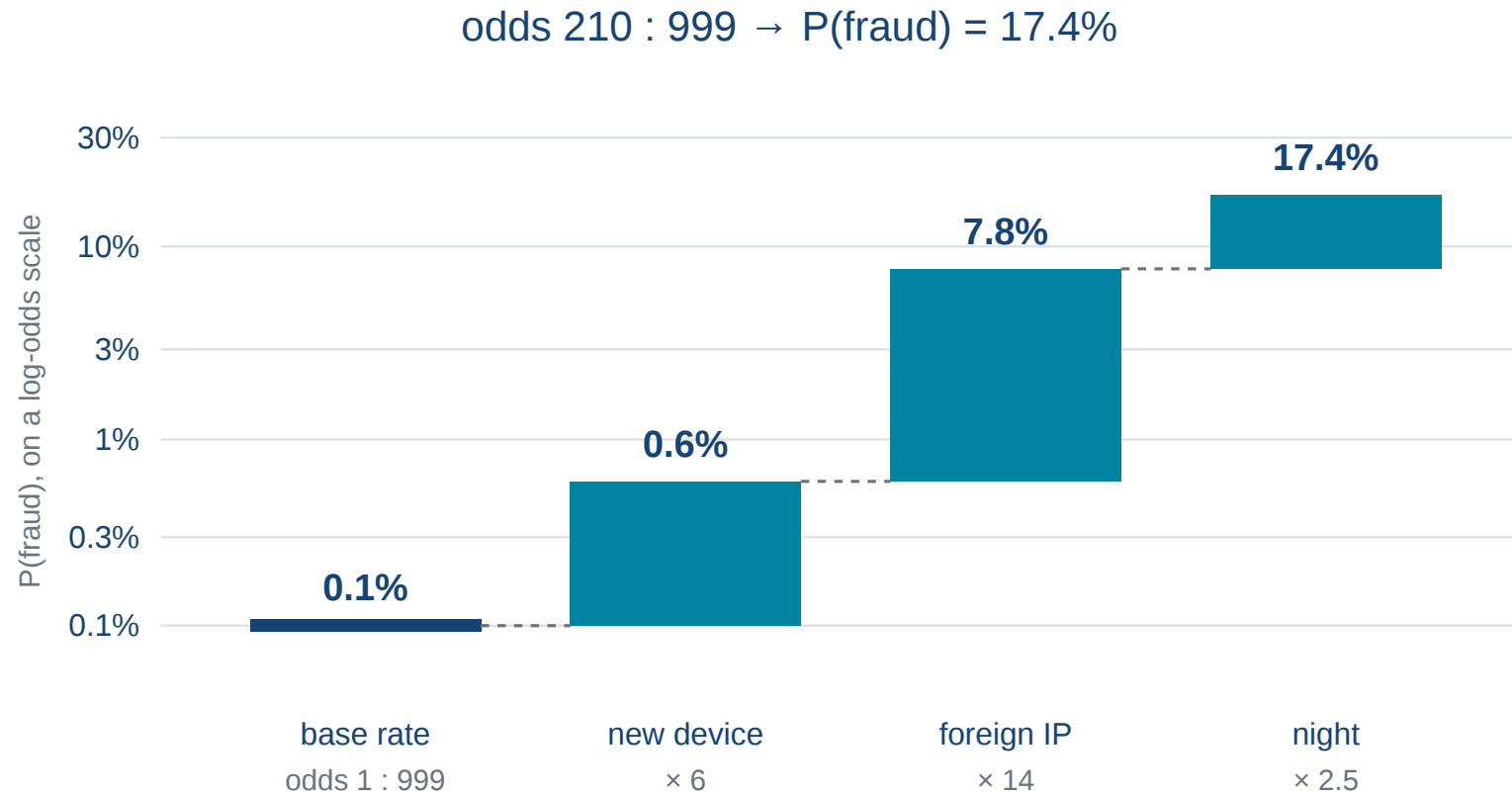
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A transaction shows all three flags. Is it more likely fraud than not?

$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \implies P(\text{fraud} \mid x) = \frac{210}{1,209} \approx 17\%$$

Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up



Predict before each click: which flag moves the probability most?

One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

KEEP THE PEAK, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — the three heads in three flips of Lecture 03, in a single cell.

Smoothing Is a Prior

Laplace smoothing adds one pseudo-count to every value of attribute j (k_j values):

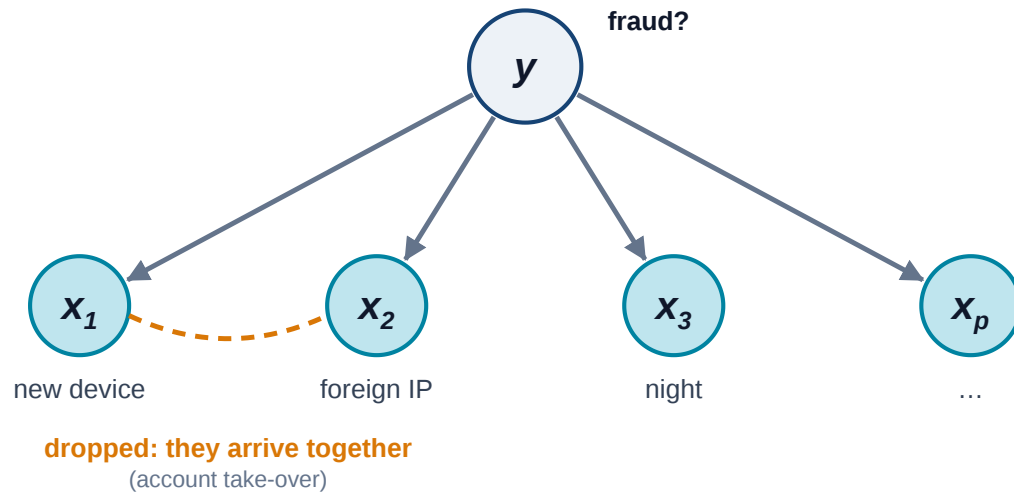
$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v, y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$ — the **posterior mean** $\frac{a+h}{a+b+n}$ of Lecture 02 under the flat prior Beta(1, 1). For $k_j > 2$ values, the flat Dirichlet does the same.

count among 100 frauds	MLE (peak)	smoothed (mean)
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

Under the flat prior the peak of 0 of 100 is still 0. Smoothing quietly **leaves the peak for the mean** — exactly where the peak was wrong.

Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ with night
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

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Up — from 17% to about 75%. The odds get a second factor 14: 2,940 : 999. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and **keeps the peak** of every count; smoothed, it keeps the Beta posterior *mean* instead. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.

Generative vs Discriminative

Guiding Question: Naive Bayes and logistic regression see the same fraud table. Which would you trust with 50 labelled frauds — and which with 50,000?

The Same Line, Fitted Twice

Take the logarithm of the naive-Bayes odds. For yes/no flags $x_j \in \{0, 1\}$ with $\theta_{jy} = P(x_j = 1 \mid y)$:

$$\log \frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)} = w_0 + \sum_{j=1}^p w_j x_j, \quad w_j = \log \frac{\theta_{j1}}{\theta_{j0}} - \log \frac{1 - \theta_{j1}}{1 - \theta_{j0}}$$

Where have you seen a log-odds that is linear in x ?

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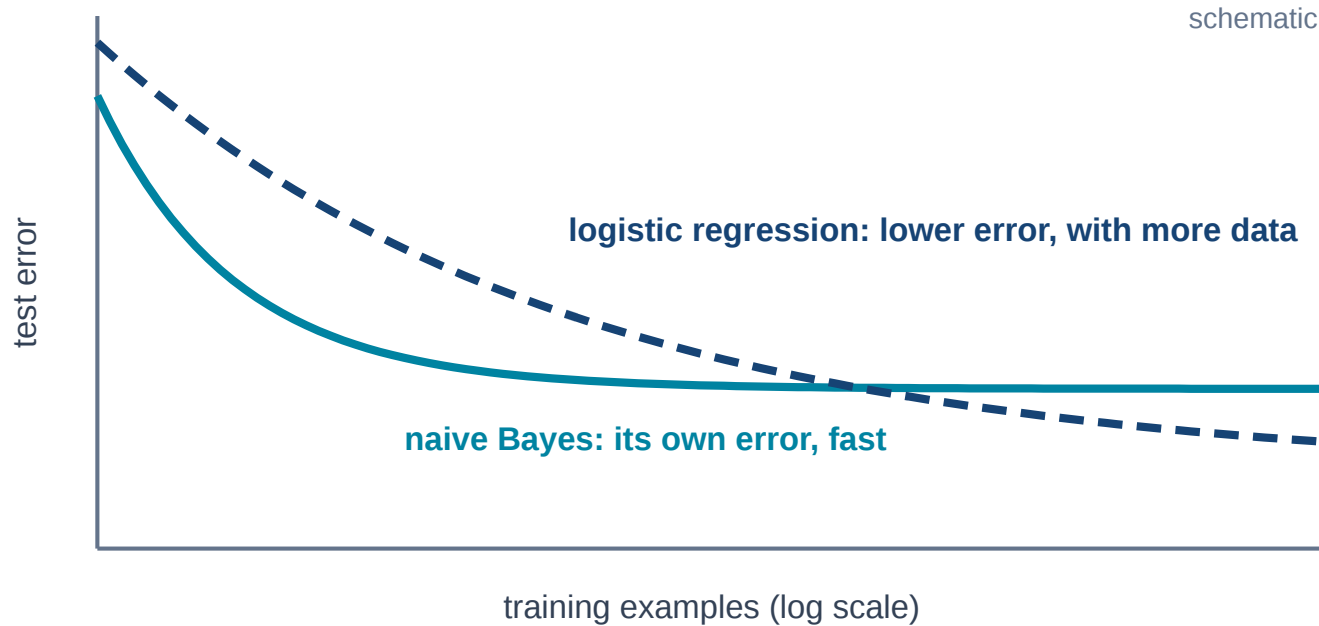
Where have you seen a log-odds that is linear in x ?

Logistic regression (Lecture 03): $P(\text{fraud} \mid x) = \sigma(w_0 + w^\top x)$. Same family of classifiers — naive Bayes sets the weights by **counting**, one attribute at a time; logistic regression by **maximising** $\sum_i \log p(y_i \mid x_i)$, all weights jointly.

Two Ways to a Classifier

	naive Bayes — generative	logistic regression — discriminative
models	$P(y)$ and $p(x y)$	$P(y x)$ only
fitted by	counting (joint likelihood)	iterative optimisation (conditional likelihood)
correlated flags	counted twice	credit shared
a missing field	drop its factor	needs a value
an unusual x	low $p(x)$: can be flagged	still a confident score
a new base rate	change $P(y)$	refit or correct the intercept
drops	dependences between attributes	everything about x

Few Data, Many Data



Few labelled frauds: naive Bayes. Its assumption acts like a strong prior — little to estimate, so it settles fast.

Many labelled frauds: logistic regression. It pays for fewer assumptions with data, and ends lower.

Shown empirically and in theory by Ng & Jordan (2002). A wrong assumption costs accuracy in the limit and buys it early — the prior of Lecture 02, again.

Mini-Exercise

Our three-flag transaction arrives **without the IP field** — the logging failed.

What does naive Bayes do with it — and what does logistic regression do?

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What does naive Bayes do with it — and what does logistic regression do?

Naive Bayes drops the factor: $\sum_v p(x_{\text{IP}} = v \mid y) = 1$, so the odds become $6 \cdot 2.5 : 999$ — about **1.5%**. **Logistic regression** has no model of x : it needs *some* value for x_{IP} , so someone must impute one, or train a second model without the field.

Takeaway

Naive Bayes and logistic regression draw **the same kind of line**; they differ in what they model. The generative model pays with assumptions about x and gets answers about x ; the discriminative one pays with data.

lecture	shortcut	what is dropped
03	keep the peak	how sure we are
03	only $p(y \mid x)$	every question about x
04	assume it apart	every dependence between attributes

Naive Bayes is a graph with one hub. **Lecture 05** draws the others — and keeps exactly the dependences we claim.